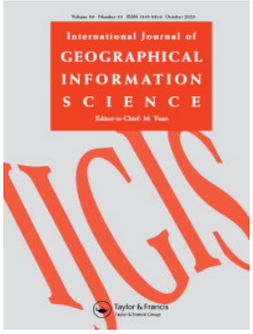

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RESEARCH ARTICLE

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The cost-balanced accessibility (CBA) model: a new framework for measuring urban accessibility with multi-objective considerations

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ABSTRACT

The two-step floating catchment area (2SFCA) model and its variants have been widely employed in spatial accessibility research. However, these models typically rely on predefined supply, demand, and travel impedance data and produce deterministic outputs, making them primarily suited for accessibility assessment rather than supporting optimization. To address this limitation, this study introduces the cost-balanced accessibility (CBA) model, an advanced framework built upon the congestion supply accessibility (CSA) model. Theoretically, the CBA model incorporates an iterative approach to eliminate instability in congestion calculations in the CSA model and establishes convergence conditions for the iterative process. Furthermore, the CBA model introduces an additional parameter to reconcile the scaling differences between travel cost and supply capacity, avoiding counterintuitive outcomes such as 'preferring distant facilities over nearby ones' in the CSA model. The CBA model also utilizes Pareto optimization to identify optimal parameters, enabling simulation of supply-demand allocation under various urban management objectives, such as maintaining average congestion or travel costs below predefined thresholds. Validated with both simulated and real-world data, the CBA model integrates multi-objective optimization into the 2SFCA framework for the first time, offering a versatile tool to adjust supply–demand balance and mobility under diverse urban management objectives.

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Spatial accessibility; supply congestion; Pareto optimization

1. Introduction

Spatial accessibility represents the chance for individuals or groups at a specific location to approach or participate in opportunities or facilities (Jones 1981). Among

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various accessibility approaches, the two-step floating catchment area (2SFCA) framework offers an explicit representation of the relationship between supply and demand in conjunction with spatial impedance (i.e. travel time or distance) (Radke and Mu 2000, Luo and Wang 2003, Tao *et al.* 2014). Owing to the favorable operability of the 2SFCA framework, it has been widely applied in accessibility research across various domains, such as healthcare, urban green spaces, education, and job opportunities (Dai *et al.* 2018, Wang *et al.* 2021, Zhang *et al.* 2022, Chen *et al.* 2024).

The improvement of the 2SFCA framework represents a central focus in current research. The prevailing improvements can be generally categorized into two types: The first category addresses the parameter uncertainty in the 2SFCA model by incorporating real-world data, such as actual population movement data, transportation modes, and service status of supply sites, to refine the model parameters accordingly (Guo *et al.* 2019, Li *et al.* 2021, Chen *et al.* 2022, Kim and Kwon 2022, Zhang *et al.* 2022, 2024, Wu and Zheng 2023); the other category targets the theoretical flaws in the 2SFCA framework, such as the fixed radius in the decay function, the neglect of competition mechanisms, and the overestimation of supply and demand. Different variants, such as the E2SFCA, 3SFCA, and Huff-2SFCA, were proposed, which have significantly advanced the theoretical development of the 2SFCA framework (Luo and Qi 2009, Dai 2010, Wan *et al.* 2012, Luo 2014, Paez *et al.* 2019). However, the 2SFCA model and most of its extensions rely on predefined inputs for supply, demand, and travel impedance (e.g. distance or time) (Tao and Cheng 2019, Gong *et al.* 2021). Once these parameters are specified, the model generates deterministic accessibility metrics. While this rigidity makes 2SFCA family of models effective for evaluative purposes in identifying spatial accessibility patterns, the static design limits its applicability in planning and decision-making, where dynamic simulation of evolving supply–demand interactions and strategic optimization of resource allocation are essential (Saxon and Snow 2020).

The congestion supply accessibility (CSA) model, which was recently developed by Lin and Cromley (2022, 2023), has shown potential in addressing the aforementioned challenges. Compared to the traditional 2SFCA model, the CSA model innovatively integrates 2SFCA and inverted 2SFCA (i2SFCA) into a unified framework, incorporating facility congestion as an additional factor into the classic 2SFCA model. This integration not only enhances the interpretive capacity of CSA model but also provides a more solid theoretical foundation than traditional 2SFCA family of models. More importantly, an additional parameter λ was introduced in the CSA model to quantify the suppressive effects of facility congestion on the attractiveness of supply points. This additional parameter λ enables the CSA model to flexibly adjust demand allocation patterns (i.e. potential population flows) by modulating the impact of facility congestion, making it highly relevant for urban management and optimization: In practice, reshaping population movement may offer greater operational flexibility and lower costs to improve the efficiency of public services compared to the renewal of urban infrastructure like public facilities or transportation network (Li *et al.* 2022, Pisoni *et al.* 2022). Through this approach, authorities could expediently achieve their predefined urban management goals such as reducing congestion and travel costs by encouraging people towards preferential routes (Abdelfattah *et al.* 2022, Teixeira *et al.* 2024).

Despite these advantages, the CSA model still faces theoretical limitations. For instance, congestion calculations can be unstable, and dimensional inconsistencies between supply capacity and travel cost may lead to counterintuitive outcomes, such as assigning higher accessibility to more distant facilities than to closer ones. Moreover, the value of the parameter λ is inherently subjective and exhibits significant uncertainty. These issues related to the CSA model will be discussed in detail in Section 2.

To address the aforementioned problems, this study introduces the cost-balanced accessibility (CBA) model by refining the CSA framework. The CBA model develops an iterative process to address the instability in congestion calculations in the CSA model and establishes the theoretical conditions for its convergence. It also introduces an additional parameter β to balance the difference between supply levels and travel costs to avoid unrealistic assumptions such as 'favoring the distant over the near'. Furthermore, the CBA model introduces the Pareto optimization method to determine the optimal combination of parameters. This facilitates the rapid simulation of accessibility outcomes and population flow patterns in alignment with predefined urban management objectives (e.g. reducing travel cost or congestion), thereby offering direct recommendations for the optimization of urban resource allocation. The effectiveness of the CBA model is demonstrated through simulated data and a case study on the accessibility of electric vehicle (EV) charging stations in Shenzhen's Luohu district. The outcome of this study will refine the existing accessibility theory and strengthen the application of the 2SFCA framework in the decision-making processes of urban management.

The remainder of this paper is organized as follows: Section 2 reviews the evolution of the 2SFCA family of models. Section 3 elaborates on the flaws of CSA and our solutions with CBA. Sections 4 and 5 present the experiments using simulation data and real-world data, respectively. Section 6 discusses and concludes this study.

2. Review on the evolution of the 2SFCA family of models

The 2SFCA offers a nuanced framework by considering both the ease of access to public services and their potential utilization by residents (Luo and Wang 2003, Dai 2011, Zhang *et al.* 2023). The 2SFCA family of models share certain variables in their definitions: S_j represents the supply at facility j, D_i denotes the demand at point i, and c_{ij} refers to the travel cost defined between them. Based on these definitions, accessibility at demand point i is quantified as the sum of the supply-to-demand ratios across all qualified facilities within the catchment area of each demand point, which can be expressed and simplied as:

$$A_i = \sum_{c_{ij}<T} \frac{S_j}{\sum_{c_{rj}<T} D_r} = \sum_j \frac{S_j}{\sum_r D_r} \tag{1}$$

where T refers to the distance or time threshold used for constructing the catchment areas.

Considering that people's willingness to visit does not abruptly change at the edge of the catchment area, previous studies have proposed numerous distance decay

functions to replace the binary boundary of the catchment area. A substantial body of literature exists in this domain, including the enhanced 2SFCA model employing a multiple discrete function (Luo and Qi 2009), the Gaussian 2SFCA model utilizing a Gaussian function (Dai 2010), and the generalized flow-based 2SFCA that incorporates adaptive power functions (Chen *et al.* 2024). These models can be unified into the following expression:

$$A_i = \sum_j \frac{S_j \times f_{ij}}{\sum_r D_r \times f_{rj}} \tag{2}$$

where f_{ij} is the impedance function of the travel cost d between location i and j.

Equation (2), with continuous change of distance decay effect, is theoretically more aligned with the patterns of human travel behavior (Wang *et al.* 2021). However, a common issue with such models is that, regardless of the number of available supply points surrounding a demand point i, the population flow to the supply point j is always fixed at $D_i \times f_{ij}$. In reality, the demand for a specific supply point should decrease as the number of supply points increases. This problem is partially solved by the 3SFCA model (Wan *et al.* 2012), which introduces an additional selection weight $G_{ij} = \frac{f_{ij}}{\sum_k f_{ik}}$ to refine the allocation of demand to supply points. Accordingly, Equation (2) could be rewritten as:

$$A_i = \sum_j \frac{S_j \times f_{ij} \times G_{ij}}{\sum_r D_r \times f_{rj} \times G_{rj}} \tag{3}$$

The inclusion of the selection weight G ensures that the demand could be proportionally distributed among all supply points. Similar models include the Huff-based 2SFCA (Huff-2SFCA) proposed by Luo (2014), in which the selection weight is assumed to be additionally affected by the attractiveness of supply points, and thus the weight G is revised as $G_{ij} = \frac{S_j \times f_{ij}}{\sum_k S_k \times f_{ik}}$.

Nevertheless, the aforementioned models still have a perceived flaw in that the sum of the denominator $\sum_r D_r \times f_{rj} \times G_{rj}$ across all supply points would always be smaller than the total population D. In other words, the total demand is deflated. The key reason for this 'demand deflation' issue lies in the decay effect of f, which essentially assumes that a certain fraction of the population will not demand the supply. Paez *et al.* (2019) addressed this issue by directly removing the distance decay factor f and rewrote the Equation (3) as:

$$A_i = \sum_j \frac{S_j \times W_{ij}}{\sum_r D_r \times W_{rj}}. \tag{4}$$

Here, the definition of W is identical to the selection weight G in the 3SFCA, i.e. $W_{ij} = \frac{f_{ij}}{\sum_k f_{ik}}$. Although this modification ensures that the total demand matches the actual one, the construction of the weight W is somewhat subjective and lacks theoretical proof.

The recent introduction of the inverted 2SFCA (i2SFCA) by Wang (2021) extends the 2SFCA framework by reversing the roles of supply and demand. Similar to the

calculation of 2SFCA, i2SFCA swaps the positions of supply S and demand D in Equation (2), which could be written as:

$$M_j = \sum_i \frac{D_i \times f_{ij}}{\sum_r S_r \times f_{ir}} \tag{5}$$

where the outcome of i2SFCA, M_j, is claimed to represent the potential congestion of the supply points (Wang 2021). Although Wang (2021) noted that the accessibility index of 2SFCA and the congestion index of i2SFCA are 'two sides of the same coin', he still treated them as independent models with different focuses. In response, Lin and Cromley (2022) revised the term W in Equation (4) by adopting the Huff probability term implied in i2SFCA (Wang 2018). Importantly, they demonstrated that this probability can be formally derived from the production-constrained spatial interaction framework (Wilson 1971), and thus proposing the ABFCA model. The ABFCA model can be expressed as the same structure in Equation (3), with W revised as:

$$W_{ij} = \frac{S_j \times f_{ij}}{\sum_j S_j \times f_{ij}}. \tag{6}$$

To further consider the role of facility congestion in affecting population flow, Lin and Cromley (2023) incorporated the congestion term M_j to adjust the supply S_j, and proposed the CSA model. The adjusted supply S_j' used in the CSA model was defined as S_j/M_j^{λ}, where λ is a parameter that indicates the intensity of congestion in discounting supply. Compared to the previously reviewed 2SFCA family of models, CSA unifies the narratives of 2SFCA and i2SFCA with further consideration of the mutual impact of congestion and the attractiveness at facilities. More details on the CSA model could be found in the original work by Lin and Cromley (2023).

However, the CSA model has at least two shortcomings. First, by replacing S_j with S_j/M_j^{λ}, CSA model necessitates an iterative process for the determination of congestion M_j according to Equations (5) and (6). This means that, in CSA, there can be discrepancies between the congestion level calculated under initial conditions and the one calculated based on the population flow size after the correction. Second, the foundation of CSA is that the Huff probability for population flow between demand point i and supply point j can be expressed as Equation (6). However, the dimensions of S and f are not consistent: while f, being related to distance, usually varies within a specific range, S can exhibit much greater variability. In extreme scenarios, a disproportionately large S_j can cause W_{ij} to approach 1, implying that people will entire population would favor facility j regardless of distance, which is unrealistic.

In addition, although the CSA model offers a parameter λ on the congestion term for adjustment, this parameter is mainly determined through subjective selection. As a result, like the 2SFCA model, CSA provides a snapshot of accessibility under fixed assumptions, yet it cannot simulate how population flows would respond to changes in planning goals, such as reducing overall congestion. To address this limitation, researchers have increasingly explored optimization-based approaches to accessibility modeling. with the Rational Agent Accessibility Model (RAAM) epresenting a notable example (Saxon and Snow 2020). The RAAM seeks to optimize the allocation of

individuals from each demand site by minimizing and balancing travel costs, defined as the sum of congestion and travel time:

$$\text{Total Cost} = \frac{\sum_i X_{ij}/S_j}{\lambda} + \frac{c_{ij}}{\tau} = \frac{\sum_i X_{ij}}{\lambda S_j} + \frac{c_{ij}}{\tau} \tag{7}$$

where X_{ij} represents the flow of people from population grid i to supply site j; c_{ij} is the travel cost; and S_j denotes the service capacity at supply site j. By fixing λ (typically set to the population-to-capacity ratio for the system), the parameter τ becomes the sole free parameter required to investigate the balance between congestion and travel cost. RAAM then assigns patients to the nearest hospital and iteratively reallocates demand from the most congested or costly hospitals to less burdened alternatives until equilibrium is reached. Despite this dynamic reallocation, RAAM has notable limitations. It requires manual tuning of key parameters, and its additive objective of minimizing congestion and travel time lacks theoretical or behavioral justification, leaving the real-world interpretation of these adjustments unclear.

3. Methodology

3.1. Convergence of congestion

CSA calculates the potential congestion based on the assumption of the production constraint from i2SFCA (Wilson 1971, Lin and Cromley 2022, 2023). For a production-constrained case, T_{ij}, which represents the flow from demand point i to supply point j, should satisfy:

$$\sum_j T_{ij} = D_i \tag{8}$$

According to the narrative of 2SFCA discussed in (Lin and Cromley 2022), Equation (8) can be further extended as:

$$\sum_j T_{ij} = D_i = \sum_j D_i \times A_i \times S_j \times f_{ij} \tag{9}$$

Then, by solving for A_i, it is easy to obtain, ED_j, the estimated population flow to supply point j, as:

$$ED_j = \sum_i T_{ij} = \sum_i \frac{D_i \times S_j \times f_{ij}}{\sum_j S_j \times f_{ij}} \tag{10}$$

Thus, the potential congestion M_j could be computed by dividing ED_j with S_j, as shown in Equation (11), which is identical to Equation (5).

$$M_j = \frac{ED_j}{S_j} = \frac{\sum_i (W_{ij} \times D_i)}{S_j} \tag{11}$$

However, when the calculation of W_{ij} involves M_j, that is, $W_{ij} = (S_j/M_j^{\lambda} \times f_{ij}) / \sum_j (S_j/M_j^{\lambda} \times f_{ij})$, the calculation of M_j becomes iterative, with the value from the previous round directly influencing the calculation of M_j in the next round. Therefore,

in the CBA model, this process is expressed as:

$$\begin{cases} M_j^{T+1} = \dfrac{\sum_i (W_{ij}^T \times D_i)}{S_j} \\ W_{ij}^T = \dfrac{S_j/(M_j^T)^\lambda \times f_{ij}}{\sum_j (S_j/(M_j^T)^\lambda \times f_{ij})} \\ M_j^0 = \sum_i \dfrac{D_i \times f_{ij}}{\sum_r S_r \times f_{ir}} \end{cases} \tag{12}$$

where the superscript T indicates the round of iteration. Notably that, in the CSA model, only M_j^0 was used, while the potential discrepancy between M_j^T and M_j^{T+1} were ignored.

It is worth noting that Equation (12) yields a stable solution only if M reaches convergence. Here, we could theoretically prove that, when f_{ij} are not all equal, the vector of M will converge when $\lambda < 1$ and diverges when $\lambda \geq 1$. The detailed proof of the convergence condition of M are referred to in the appendix.

3.2. Reconcile dimensional disparity of supply S and impedance function f

According to Equation (12), it is easy to know that, when a certain S is very large, its congestion level will also be relatively low, making the corresponding W to approach 1. This will lead almost all the population to be attracted to this S point while ignoring the actual travel cost. However, this situation goes against the natural law of people making rational choices.

To address the problem, we added an exponent factor β to f_{ij} in the calculation of W to reconcile the dimensional disparity between S and f, where β determines the intensity of travel cost in shaping people's visitation behavior. As a result, the term W_{ij} will be rewritten as:

$$W_{ij}^T = \frac{S_j/(M_j^T)^\lambda \times (f_{ij})^\beta}{\sum_j (S_j/(M_j^T)^\lambda \times (f_{ij})^\beta)} \tag{13}$$

Building upon the previous discussions, we hereby propose the CBA model, which is expressed as:

$$A_i = \sum_j \frac{S_j \times \hat{W}_{ij}}{\sum_i (D_i \times \hat{W}_{ij})} \tag{14}$$

where $\hat{W}_{ij}$ represents the value of W_{ij} when it converges in the following iterative process:

$$\begin{cases} W_{ij}^T = \dfrac{S_j/(M_j^T)^\lambda \times (f_{ij})^\beta}{\sum_j (S_j/(M_j^T)^\lambda \times (f_{ij})^\beta)} \\ M_j^{T+1} = \dfrac{\sum_i (W_{ij}^T \times D_i)}{S_j} \\ M_j^0 = \sum_i \dfrac{D_i \times (f_{ij})^\beta}{\sum_r S_r \times (f_{ij})^\beta} \end{cases} \tag{15}$$

Specially, according to the above equation, the average travel cost $\overline{d}$ for the entire population could be calculated as:

$$\overline{d} = \frac{\sum_{i,j}\left(d_{ij} \times \hat{W}_{ij} \times D_i\right)}{\sum_{i,j}\left(\hat{W}_{ij} \times D_i\right)} \tag{16}$$

where $\sum_{i,j}(\hat{W}_{ij} \times D_i)$ represents the total population, and $\sum_{i,j}(d_{ij} \times \hat{W}_{ij} \times D_i)$ represents the total travel cost between all demand-supply pairs.

3.3. Selecting λ and β using the Pareto optimality technique

Reducing travel cost and congestion are widely acknowledged as the main tasks in planning the utilization of public facilities (Allam *et al.* 2022, Liang *et al.* 2024). According to Equation (15), there is a roughly negative correlation between travel cost (i.e. impedance function f) and congestion M, meaning that a decrease in one will lead to an increase in the other. However, for CBA, the congestion of each supply site, the travel cost for people to access them, as well as the accessibility results, are influenced by the model parameters λ and β. Consequently, the challenge is to identify an optimal set of parameters for the CBA model that ensures simulated indicators align as closely as possible with a given urban management goal (e.g. controlling congestion or travel cost), while simultaneously maintaining near-optimal values for other relevant metrics.

To address the aforementioned issue, we employ the Pareto optimization method for the selection of model parameters. Pareto optimality refers to a solution strategy whereby, in the absence of further improvements in one objective, efforts are directed toward maximizing the value of other objectives (Censor 1977). The fundamental principle of Pareto optimality lies in achieving a balance among conflicting objectives, striving to enhance other objectives as much as possible when one objective cannot be further improved under current conditions, thereby attaining a relatively optimal solution.

To determine parameters λ and β, it is necessary to specify the constraints that must be satisfied during the optimization process. Considering the characteristics of the CBA model and practical goals in urban management, these constraints could include but are not limited to the restriction on the average travel cost $\overline{d}$ or congestion level $\overline{M}$. Therefore, given a special urban management goal/constraint G and its upper limit of T, the parameter estimation can be cast as a multi-objective optimization problem:

$$\min_{\lambda,\beta} \; f(\lambda,\beta) = \begin{bmatrix} \overline{d}\ (\lambda,\beta) \\ \overline{M}\ (\lambda,\beta) \end{bmatrix}, \; s.t.\; G(\lambda,\beta) \le T. \tag{17}$$

Subsequently, a grid search approach is employed to calculate $\overline{d}$ and $\overline{M}$ under different parameter combinations of λ and β. Lastly, a Pareto front is derived from the grid search results, and the optimal parameter set is selected subject to the predefined constraints.

The parameter λ and β are identified through a Pareto optimization framework that simultaneously considers two competing objectives: minimizing the average travel

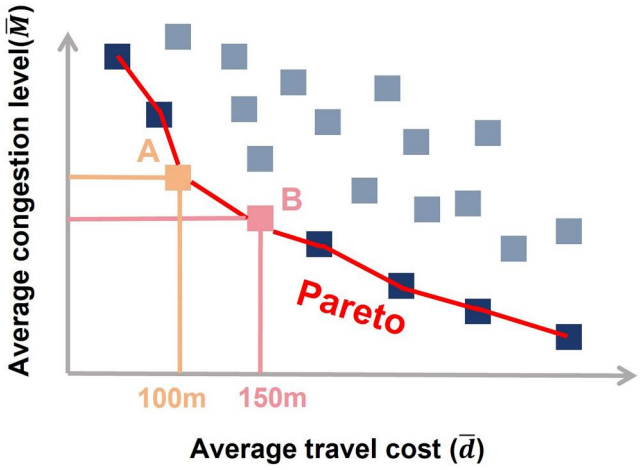

Figure 1. Illustration of Pareto optimization used in the CBA model.

cost $\overline{d}$ and minimizing the average congestion level $\overline{M}$. For each candidate (λ, β) pair in the grid search, the corresponding values of $\overline{d}$ and $\overline{M}$ are computed to form the outcome space. A parameter pair is considered Pareto optimal if there exists no other combination that achieves a lower value in one objective without increasing the other. Figure 1 illustrates an example of the above process. Each combination corresponds to one point in the outcome space, and the red line connects these non-dominated solutions and forms the Pareto front. The choice of final model parameters depends on the planning constraints. For example, Point A represents the solution that minimizes congestion subject to a 100 m travel cost limit, whereas Point B reflects the solution with a relaxed cost tolerance of 150 m. Once an optimal solution is selected from the Pareto front based on the planning goal, the corresponding λ and β are applied in the CBA model to compute accessibility metrics.

4. Simulation experiments

In this section, we conducted an experiment using a simulated dataset to justify the improvements in the proposed CBA model compared to CSA model and demonstrate its effectiveness through a comparison with two representative accessibility models.

4.1. Convergence of M with the iterative process in CBA

In the first experiment, we aimed to demonstrate the convergence of the congestion variable M_j under different values of λ. As shown in Figure 2, we constructed a simulated dataset featured with three supply points and two demand points. The capacities for supply points S_1, S_2 and S_3 are 10, 30, and 100, respectively. The distances of these supply points to demand point D_1 are 0.5, 4 and 10 km, and to demand point D_2 are 0.6, 1 and 1.2 km, respectively. Specifically, we set $\beta = 1$ and used the Gaussian-based function $f_{ij}(d) = e^{-\frac{d^2}{2\sigma^2}}$, where d represents distance and σ is consistently set to 5 in the simulation experiments.

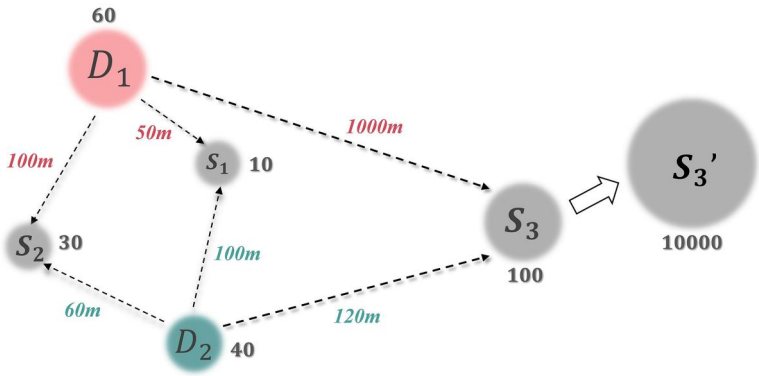

Figure 2. Diagram for the simulated data.

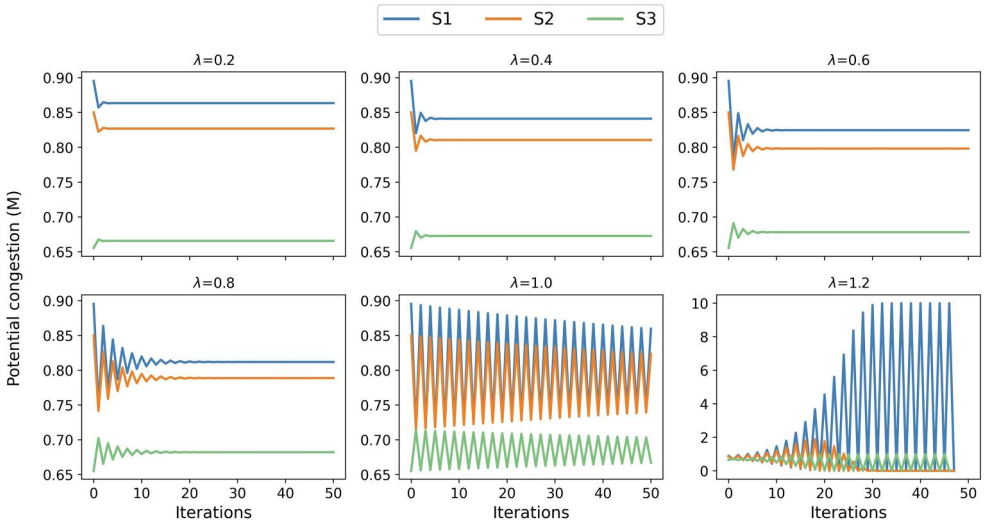

Figure 3. Iteration process of M for different λ.

Figure 3 illustrates the change of potential congestion M with different values of λ with 50 iterations. Notably, the convergence criterion was set to $\epsilon = 10^{-6}$, where ϵ represents the threshold for the change of M values between iterations. Some observations can be made as follows:

- When $\lambda < 1$, We observe that M generally converges after several iterations, with smaller λ values requiring fewer rounds to reach convergence;
- When $\lambda = 1$, all M values exhibit pronounced oscillations, alternating between distinct congestion levels;
- When $\lambda > 1$, M_{S1} is rising rapidly while M_{S2} seems to be equal to 0. The overall results exhibit a divergent pattern.

The above observation aligns with our expectations and provides empirical evidence to substantiate the correctness of our theoretical derivation regarding the convergence criteria of the CBA model.

4.2. Effect of parameter β

As mentioned in Section 3.2, when the dimensional issue between supply and travel cost is neglected, the model described by Equation (12) will allocate most people to the supply point with a very high supply volume regardless of the travel costs.

To elucidate this issue, we modified the simulation case presented in Section 4.1 by increasing the supply capacity of the remotest supply point S_3 to 10,000, as shown in Figure 4. We also fixed $\lambda = 1$ and ignored the iterative calculation of M intentionally to focus on the impact of β. Specially, we calculated (i) the proportion of the total population allocated to each supply point, defined as $\sum_i W_{ij}D_i/D$ following Equation (12), and (ii) the average travel distance $\bar{d}$ for the evaluative purpose.

As depicted in Table 1, when the dimensional issue between f and S is ignored (e.g. β is set to 1), over 98% of the potential demand is allocated to $S3$. This indicates that the vast majority of people are opting for a more distant supply point over a closer one because of its larger supply capacity.

The irrationality behind the above behavior could be further explained with a small tale: when you interview random residents if they prefer to spend 5 min to a nearby

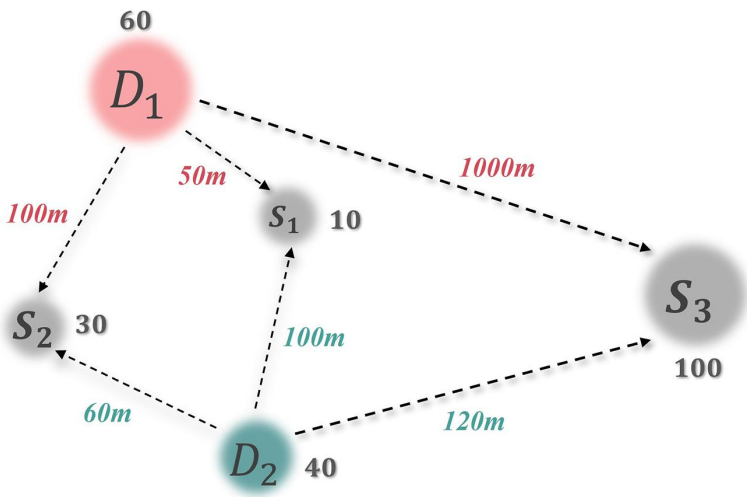

Figure 4. Simulated data after modifying the supply of S_3.

Table 1. Effect of β on the average travel cost and proportion of total population received by each supply site.

β	S_1	S_2	S_3'	$\bar{d}$ (km)
0.5	0.20%	0.53%	99.27%	6.44
1	0.47%	1.07%	98.46%	6.38
1.5	1.17%	2.24%	96.59%	6.24
2	2.88%	4.67%	92.45%	5.94
2.5	6.57%	9.04%	84.39%	5.32
3	12.86%	15.07%	72.08%	4.37
3.5	20.56%	20.57%	58.87%	3.30
4	27.40%	23.41%	49.19%	2.48
4.5	32.42%	23.67%	43.91%	1.99

medium-sized medical facility or 2 h to a distant super-large medical center, it could be expected that most people would opt for the former. This is because, for individuals, travel time, or distance, is an explicit cost, while the increase in the scale of the supply site is an implicit benefit (Geurs and Van Wee 2004, Litman 2008). Consequently, in the vast majority of cases, travel costs are the primary consideration in people's travel decisions, and the situation is that over 98% of people visit S_3 will not happen in reality.

The issues encountered above were significantly improved after increasing β. As β increases, the impact of the supply capacity on people's choice of supply points gradually diminishes, with more demand being assigned to the closer supply points. This results in a substantial reduction in the average travel distance, and the population received at each supply point becomes more evenly distributed. In practical applications, the parameter β helps mitigates counterfactual results arising from unrealistic assumptions, allowing the model to be more flexible in adapting to population allocation and accessibility assessments under different requirements.

4.3. Sensitivity analysis of model variables

To gain insights into the influence of parameters λ and β on the variables of the CBA model, we evaluated the average congestion level $\overline{M}$ and the average travel cost $\overline{d}$ across various values of λ and β. As depicted in Figure 5(a), $\overline{d}$ exhibits a sharp decline in response to the increase in the value of β, whereas it trends upwards slightly with the increase of λ. In contrast, as shown in Figure 5(b), $\overline{M}$ surges as β grows, and declines significantly when λ increases. The paradoxical interplay between congestion and travel costs underscores the need for a Pareto optimization framework in CBA, which enables a balanced treatment of these competing objectives. Additionally, the influence of λ on the model outcomes is rather limited when β is fixed at 1 (equivalent to the CSA model), whereas incorporating β broadens the parameter space and improves the scalability and adaptability of the CBA model.

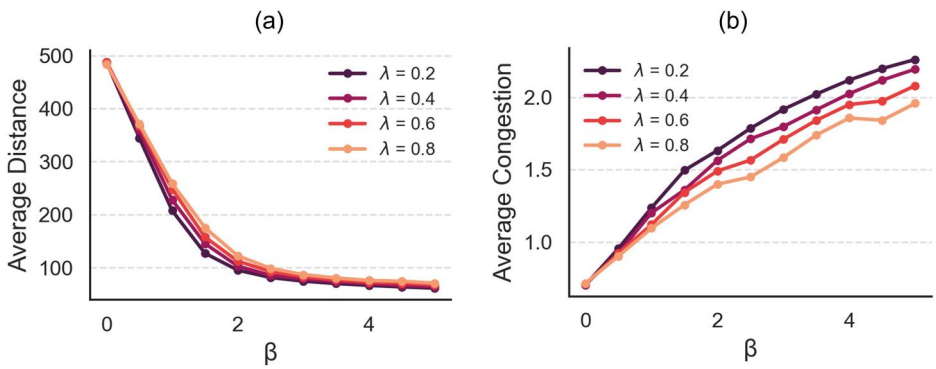

Figure 5. Curves of average travel cost $\overline{d}$ and average congestion $\overline{M}$ under different λ and β settings. (a) $\overline{d}$ decreases with λ and β; (b) $\overline{M}$ decreases with λ and β.

5. Case study: accessibility to EV charging stations in Luohu District, Shenzhen

5.1. Study area and data description

5.1.1. Study area

Shenzhen, China, is at the forefront of EV charging infrastructure construction both nationally and globally. However, due to the large population and complexity of activities in Shenzhen, optimizing the distribution and utilization efficiency of EV charging stations has always been a significant challenge in the city's management. Luohu District, as shown in Figure 6, was located in the central part of Shenzhen and was chosen as the study area for this paper. As reported in the Shenzhen Statistical Yearbook 2023,[1] Luohu District exhibits the third highest population density within the city and significantly surpasses the average density of Shenzhen. Simultaneously, as one of the earliest developed regions in Shenzhen, Luohu District exhibits a complex internal building and habitat environment, with the issue of uneven infrastructure distribution and challenges in city renewal being particularly pronounced. Given these facts, analyzing the accessibility of EV charging stations in Luohu District could be a typical instance for guiding future development and optimization of charging infrastructure in the area.

5.1.2. EV charging stations data

Data on EV charging stations in Luohu District were obtained from the Shenzhen Municipal Government Data Open Platform.[2] The data was last updated in September 2023. We used the Amap, which is one of the largest providers of map service in China, to search for the address of each charging station and obtain its accurate spatial locations for further analysis.

Considering the difference in power supply efficiency between fast and slow charging piles, we used the following formula to calculate the supply capacity of each charging station:

$$EV = P_s \times N_s + P_f \times N_f \tag{17}$$

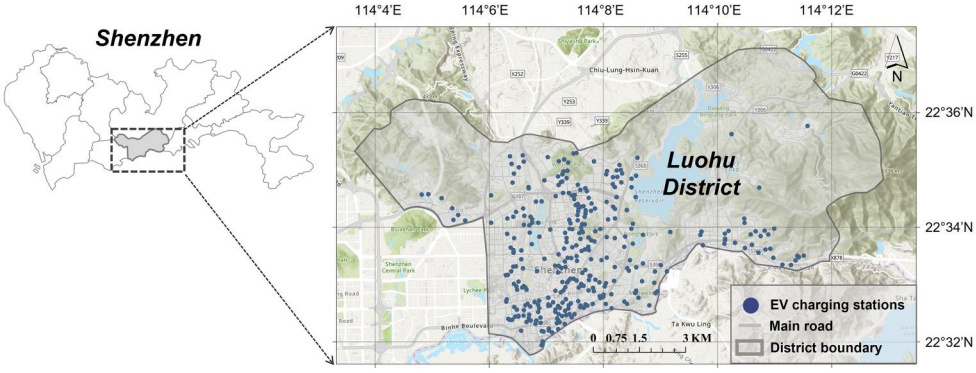

Figure 6. Map of Luohu District.

where N_s is the number of slow charging piles and N_f is the number of fast ones, P_s and P_f refer to their charging powers, respectively. According to 'Shenzhen New Energy Vehicle Charging and Swapping Facilities Management Measure'[3] and the local standard 'Electric Vehicle Supercharging Equipment Grading and Evaluation Specifications' (DB4403/T 433-2024), the output power of newly built charging facilities should not be less than 7 kW, and the power of fast charging equipment mainly ranges from 60 kW to 200 kW. In this study, we uniformly set power values P_s to 7 kW for slow charging piles and P_f to 120 kW for fast ones.

5.1.3. Mobile phone data

Mobile phone data are widely used in recording user location and have promoted a wide board of mobility studies. This research employs mobile phone data to calculate the population demand for EV charging stations. The raw mobile phone data in Shenzhen was collected from 29 May to 11 June 2023 by China Unicom. Particularly, we utilized the aggregated results of raw data on a regular grid, which allows for the provision of population counts on an hourly basis within 500 m $\times$ 500 m grid cells. It is important to note that, since China Unicom users represent only a portion of the total population, the aggregated mobile signaling data were scaled to the total Shenzhen population to ensure consistency with official statistics. It should also be noted that, as of the end of June 2023, the number of EVs in Shenzhen reached 862,600, accounting for approximately 21% of the total vehicle ownership.[4] Therefore, we estimated EV charging demand in each grid cell as 21% of the mobile phone-based population, assuming an even spatial distribution of EV ownership across the city.

5.2. Model performance evaluation and comparison in two scenarios

To demonstrate the applicability of the CBA model across different cases, we simulated two scenarios, referred to as Limited Travel Cost and Limited Congestion. The specific configurations of these scenarios are detailed in Table 2. In the scenario of Limited Travel, the travel cost constraint was set to 6 min, which corresponds to a travel distance of about 2 km according to the average driving speed reported by the Shenzhen Municipal Bureau of Transportation.[5] In the scenario of Limited Congestion, the upper limit of supply congestion was calculated as 1.35, which represents 95% of the average congestion level achieved by the 2SFCA model in this experiment.

Table 2. Scenario descriptions and parameter settings.

Scenario	Description	Experimental configuration
Limited Travel Cost	Overall travel cost is set as the key constraint to ensure that residents' travel cost remains within a fixed threshold.	The average travel cost $\bar{d}$ should not exceed 6 min.
Limited Congestion	Average congestion is set as the key constraint to ensure that wherever residents travel, congestion remains within a reasonable range.	The average congestion $\bar{M}$ should not exceed 95% of the one achieved by the classic 2SFCA model.

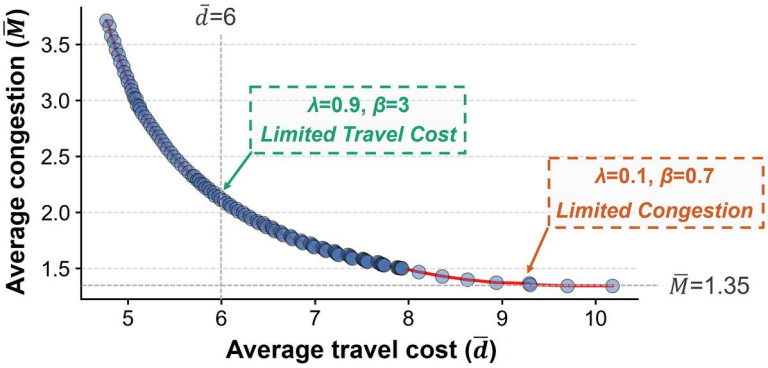

Figure 7. Optimal parameters identified for each scenario based on the Pareto frontier.

To determine the optimal parameters combination for each scenario, we performed a grid search with λ ranging from 0 to 1 and β from 0 to 5, both using a step size of 0.1. The average facility congestion $\overline{M}$ and travel cost $\overline{d}$ were calculated under different parameters to construct the Pareto frontier. Specifically, we used the Gaussian function with a threshold of 5 min to represent the distance decay effect in the following experiments. As shown in Figure 7, the optimal parameters were identified as $\lambda = 0.9$ and $\beta = 3.0$ for *Limited Travel Cost*, and $\lambda = 0.1$ and $\beta = 0.7$ for *Limited Congestion* Scenario.

5.2.1. Limited Travel Cost Scenario

To demonstrate the superiority of the CBA model, we compared it with the classic 2SFCA model and the CSA model. Specifically, the CSA model is configured with the same parameter $\lambda = 0.9$ as the CBA model for comparative analysis.

As shown in Figure 8(a), the classic 2SFCA model has the lowest average accessibility of 0.689 and the highest Gini coefficient of 0.295. Figure 8(b) shows that the CSA model achieves higher accessibility and exhibits extremely high equality with the lowest Gini coefficient of 0.075. However, the reliability of the CSA result might be questionable, as the supply-demand mismatches in charging infrastructure in Shenzhen's Luohu District have been highlighted in the government reports.[6] This suggests that the observed high level of equality might not accurately reflect real-world conditions. By comparison, the CBA model, as shown in Figure 8(c), achieves the highest average accessibility while maintaining a relatively reasonable Gini coefficient of 0.167.

Additionally, as shown in Figure 9, the CBA model significantly reduces travel time costs while introducing a certain degree of congestion. This characteristic is crucial in most time-sensitive scenarios in daily life. For example, when constructing the so-called '15-min living circle', the results of the CBA model can be used to guide individuals to public resources with lower travel costs while highlighting congested facilities, thereby informing targeted infrastructure expansion and more effective urban planning.

5.2.2. Limited Congestion Scenario

Similar to the experiments conducted for *Limited Travel Time* scenario, 2SFCA and CSA ($\lambda = 0.1$) models were also included for comparison in this analysis. While the results

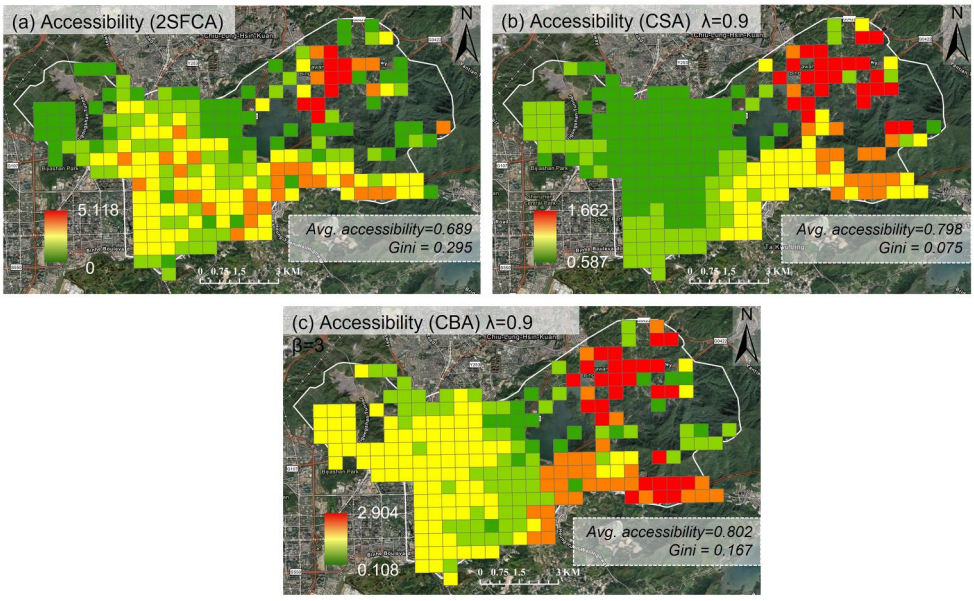

Figure 8. Spatial visualization of accessibility for different models under Limited Travel Cost Scenario.

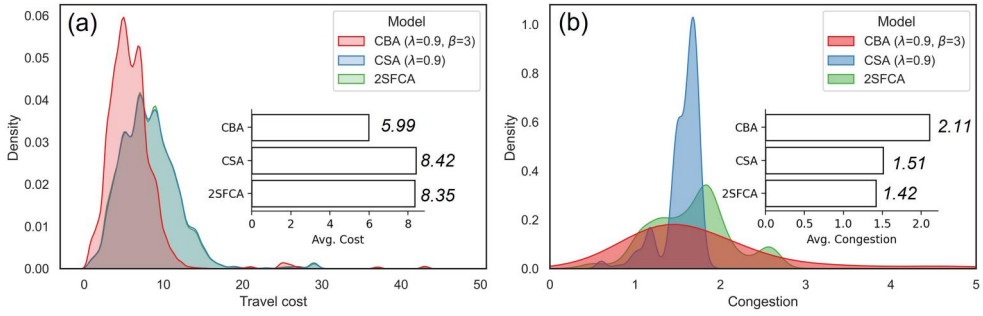

Figure 9. Distribution of travel cost and congestion under Limited Travel Cost Scenario.

of 2SFCA model, as the baseline model, remain unchanged due to its lack of adjustable parameters. Consequently, it still exhibits the lowest accessibility and the highest Gini coefficient, as shown in Figure 10(a). The accessibility results of the CSA and CBA models are relatively similar, with the CBA model yielding a slightly higher average accessibility (0.850 versus 0.835) and a minor decrease in the Gini coefficient (0.098 versus 0.103). Furthermore, as illustrated in Figure 11, the CSA model exhibits the highest average congestion of 1.43, while the CBA model significantly reduces the average congestion to 1.35, which accurately meets the requirement of this scenario.

In contrast to the previous experiment, the CBA model reduces congestion at supply points by slightly increasing travel costs. This is particularly useful for scenarios sensitive to congestion. For example, during the COVID-19 pandemic, many public facilities were closed to prevent crowd gatherings, which led to a significant reduction in public service levels. However, under the guidance of the CBA model, we can

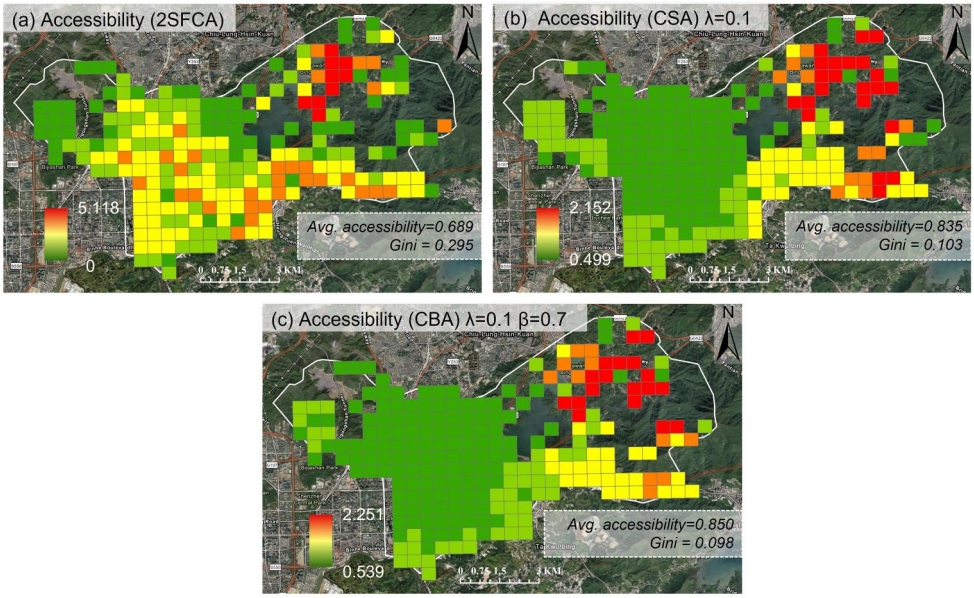

Figure 10. Spatial visualization of accessibility for the three models under Limited Congestion Scenario.

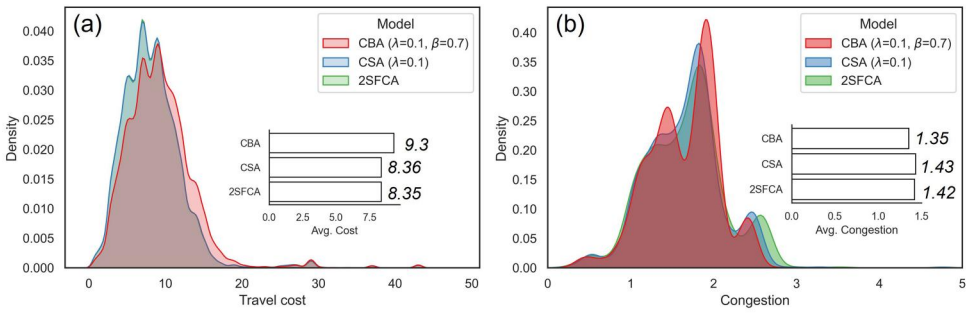

Figure 11. Distribution of travel cost and congestion under Limited Congestion Scenario.

implement targeted public facility allocation for residents (even though this may increase travel distances in some cases) to reduce the risk of crowd aggregation while maintaining the normal operation of public facilities.

5.2.3. Spatial analysis of different models

We further conducted a visual analysis of the spatial distribution of accessibility across different models and scenarios using the kernel density estimation method. For the 2SFCA model, since only travel costs are considered, people tend to choose the nearest locations with more supply to access services. This results in high accessibility concentration in areas where supply exceeds demand, forming a single-center distribution pattern, as shown in Figure 12(a). In comparison, by accounting for the attenuation effect of congestion, the CSA model, as shown in Figure 12(b,d), reallocates a portion of the population to areas with surplus supply, effectively diluting the demand on

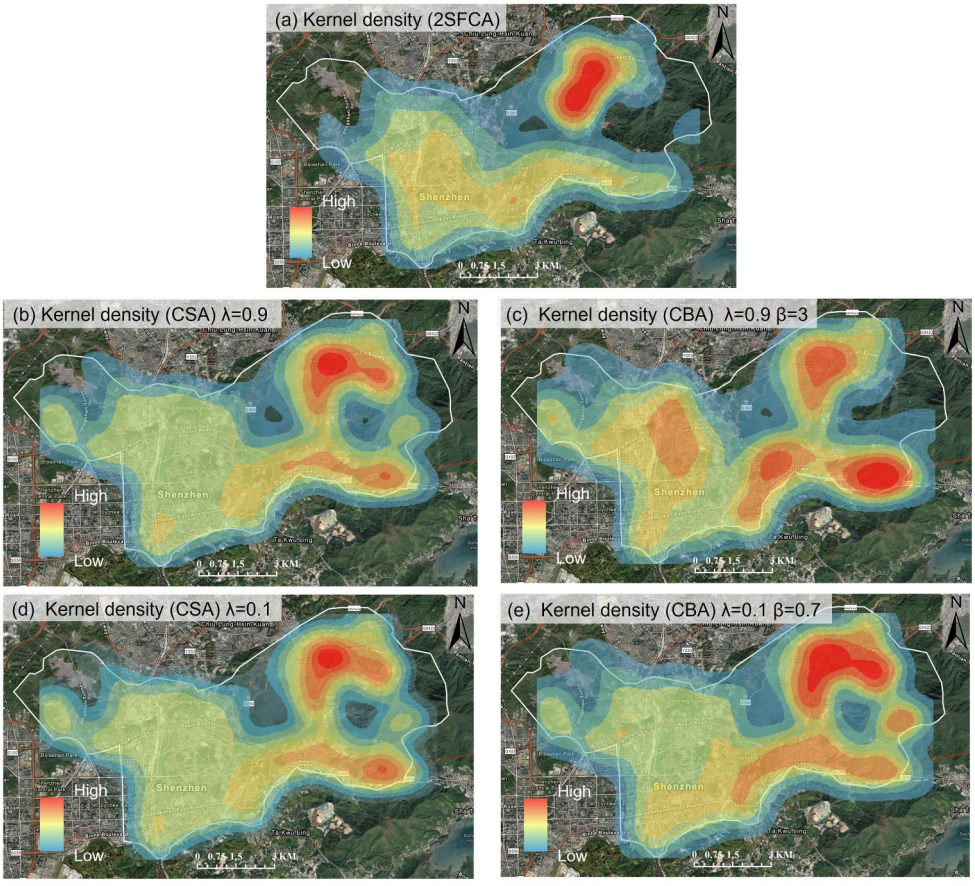

Figure 12. Spatial kernel density analysis of accessibility.

heavily utilized supply points. The CSA model partially alleviates the single-center pattern observed in the 2SFCA model, resulting in a moderated multi-center spatial distribution.

In contrast, the CBA model takes both cost and congestion into account in its travel decision-making process, leading to more rational choices when accessing resources: on the one hand, it encourages residents to moderately increase their travel costs to alleviate overcrowding; on the other hand, by optimizing population allocation, the CBA model minimizes the travel costs to public facilities within the given constraints. Consequently, the accessibility results generated by the CBA model exhibit a clear multi-center pattern, as displayed in Figure 12(c,e). This suggests that, under the CBA model, public resources are utilized more efficiently and effectively, with the spatial distribution aligning more closely with Shenzhen's multi-center urban development structure (Chen and Lai 2023).

To illustrate spatial differences among the accessibility models, we first normalized each model's outputs to the [0,1] range to ensure comparability, and then computed the absolute differences between models. Figure 13 visualizes these differences across the study area.

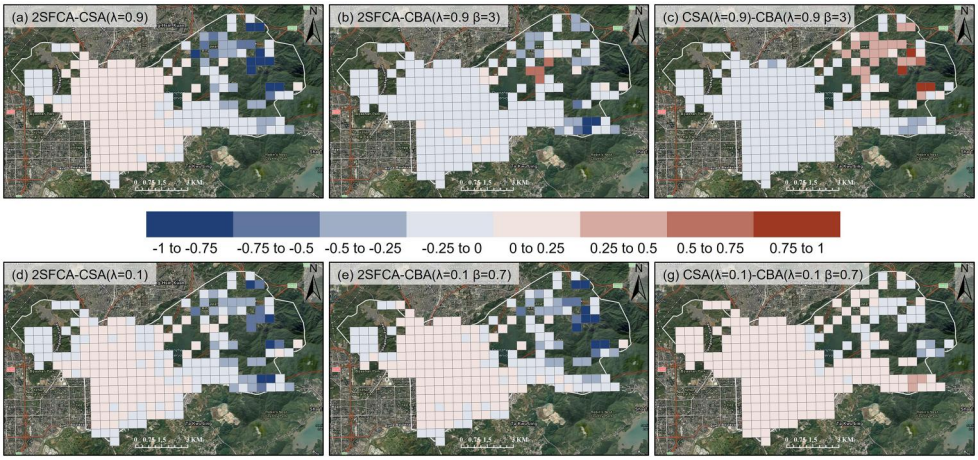

Figure 13. Normalized accessibility differences.

Under the *Limited Travel Time* scenario, the normalized accessibility values produced by the CBA model are generally higher than those of the 2SFCA and CSA models across most urban areas, particularly in the central city zones. This outcome aligns with the predefined 6-min travel time constraint: under such a constraint, residents tend to avoid long-duration trips. The CBA model, through its flow adjustment mechanism, reduces inbound travel from suburban areas, while supply remains unchanged, thereby enhancing accessibility in the city center. The differences between 2SFCA and other models exhibit fragmented spatial fluctuations, with adjacent grids showing inconsistent directions, indicating weak spatial coherence and limited interpretability. In addition, in the northeastern part of Luohu District, a suburban fringe area, CSA yields notably higher accessibility than CBA. This may be because the CSA model tends to produce more evenly distributed accessibility (see in Figure 10, Gini = 0.098), which may overstate access in peripheral locations with sparse demand or limited facility availability.

Under the *Limited Congestion* scenario, the CBA model exhibits generally lower normalized accessibility across most urban areas. This trend is a direct result of the model's mechanism: to meet system-wide congestion targets, longer-distance trips are allowed, encouraging some suburban residents to shift toward large-capacity central facilities. As supply remains constant, the inflow of suburban demand into central facilities reduces accessibility in urban areas, while peripheral zones, such as the northeastern suburbs, exhibit improved accessibility under the CBA model.

6. Discussion

6.1. Theoretical advantages of the CBA model

The CBA model retains the key strengths of classical 2SFCA variants, including the mitigation of demand deflation (Paez *et al.* 2019), the incorporation of supply-side competition (Lin and Cromley 2022), and the avoidance of arbitrary parameter settings

(Chen *et al.* 2024). Building upon this foundation, it further advances the theoretical integration of the 2SFCA and i2SFCA frameworks: in contrast to the CSA model, which introduces congestion effects without resolving their dynamic behavior, the CBA model adopts a recursive formulation and establishes formal convergence conditions. This guarantees that its outputs reflect a stable equilibrium state, thereby enhancing both reliability and interpretability.

Beyond its theoretical robustness, the CBA model introduces a tunable parameter that aligns supply capacity with travel impedance. This adjustment reduces structural distortions and improves adaptability across diverse application contexts. Moreover, through its integration with the Pareto optimization method, the CBA model reconceptualizes accessibility as an outcome shaped by competing planning objectives, rather than a static evaluative metric. As a result, the CBA model evolves into a responsive, decision-support tool for urban governance.

6.2. Practical implications of the CBA model

The CBA model holds significant potential for various political implications. Its ability to simulate and optimize supply-demand dynamics under various objectives provides policymakers with a practical tool for enhancing urban management strategies: without increasing or altering existing infrastructure, the government can make external interventions in guiding residents' visitation to certain infrastructure, thereby improving the efficiency of infrastructure usage. For example, in the case of nonrival or substitutable infrastructure such as EV charging stations, the government can implement targeted incentives to motivate residents in certain areas to utilize designated charging stations to reduce overall travel time or congestion. Furthermore, through government guidance, individuals with travel advantages (i.e. those who can obtain sufficient supply in a shorter time) can develop a consciousness of actively sacrificing personal interests, thereby achieving long-term improvements in collective welfare.

The CBA model also demonstrates better interpretability and practicality. The fluctuations and convergence properties of congestion vividly reflect the collective gaming behavior exhibited by individuals: the initial level of congestion M is determined solely by travel costs and supply capacities, reflecting individuals' expectations of congestion at supply points. However, as individuals tend to avoid highly congested areas, they naturally seek alternative, less crowded supply points. This behavior, rooted in rational decision-making, can lead to rapid overcrowding of previously underutilized locations. Over time, as population flows stabilize, congestion M reaches an equilibrium state, which represents the balance of people's comprehensive considerations of congestion, distance, and supply. In addition, compared to the CSA or RAAM model (Saxon and Snow 2020, Lin and Cromley 2022, 2023), the dual model parameters in CBA are directly determined towards the specific requirement of physical quantities (i.e. travel cost and congestion), which enhances the flexibility of CBA across different scenarios. For instance, CBA could also be applied in scenarios such as epidemic prevention and control by simultaneously controlling congestion and mobility costs (Wang 2012, Cromley and Lin 2022, Xia *et al.* 2022).

6.3. Contextual strengths of different accessibility models

It should be noted that the purpose of comparing 2SFCA, CSA, and CBA models in this study is not to establish a universal benchmark of 'better' or 'worse' performance, as is often done in deep learning or predictive modeling. As highlighted by Demitiry *et al.* (2022), different accessibility methods provide complementary perspectives and each has unique advantages and limitations, depending on the context of application. In this regard, 2SFCA remains a simple and intuitive tool for evaluating basic spatial supply-demand relationships; CSA improves upon it by incorporating congestion effects; and the CBA model further advances the CSA framework and offers greater flexibility for simulating flow adjustments under diverse urban management objectives.

6.4. Limitations and future work

This study has limitations. First, the unique characteristics of facilities such as hospitals, including their scale, function, and service quality (Jin *et al.* 2019), have not been fully considered in the current model. Future research could incorporate these aspects into the CBA framework to expand its application in a broader range of fields. Second, the current study optimizes only travel costs and congestion, which may not fully reflect the multifaceted nature of accessibility. Future work could incorporate additional indicators, such as accessibility or the Gini index, into the construction of Pareto frontiers to better accommodate diverse policy objectives. Lastly, this instant study does not account for demand dynamics. Future research could incorporate real-world population data to explore the dynamic patterns of travel costs and congestion, thereby providing more effective support for urban management.

7. Conclusion

This study introduces the CBA model as a theoretical extension of the CSA framework. While retaining the strengths of CSA, CBA incorporates context-aware parameters to resolve issues of variable scaling and demonstrates improved adaptability across diverse policy scenarios. More importantly, unlike conventional 2SFCA approaches focused on static assessment, CBA allows for flexible adjustment of supply-demand relationships in line with planning objectives, shifting accessibility modeling from descriptive analysis toward a more proactive role in spatial planning. This extension broadens the application scope of accessibility models and offers a new direction for theory-driven research.

Notes

1. https://tjj.sz.gov.cn/zwgk/zfxxgkml/tjsj/tjnj/
2. https://opendata.sz.gov.cn/
3. https://fgw.sz.gov.cn/gkmlpt/content/10/10877/post_10877362.html#25135
4. https://new.qq.com/rain/a/20230803A0A55Y00
5. http://tocc.jtys.sz.gov.cn/#/
6. https://www.szlh.gov.cn/zwgk/zcwj/content/post_10906153.html

Authors' contributions

Pengfei Chen: conceptualization, methodology, formal analysis, investigation, writing–original draft, writing–review and editing, supervision, funding acquisition. Mengjie Gong: software, validation, formal analysis, data curation, writing–original draft, writing–review and editing, visualization. Shaodong Li: data curation, writing–review and editing, supervision.

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Data and codes availability statement

The data that support the findings of this study are available in Figshare at https://doi.org/10.6084/m9.figshare.26969014.v2

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